

Packages

Reference Sheet

July 15, 2026

Copy-paste package list — paste this into your preamble, right after `\documentclass{...}`:

```
\usepackage[a4paper, margin=1in]{geometry}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{listings}
\usepackage{xcolor}
\usepackage[hidelinks]{hyperref}
```

Is this document in ACM format? No. The solution uses `\documentclass[11pt]{article}` — the standard LaTeX article class, single-column. It is *not* `acmart`. The CSE 200 paper-reproduction assignment needs `acmart`; this B1 exam reproduced a plain article, so `article` is the right class here. The two are different tasks with different format requirements.

This sheet lists every package loaded in `main.tex` and the key commands used from each, in the order they appear in the preamble. Code is shown literally; nothing here needs to compile on its own.

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1 Packages Loaded

Package	What it provides here
geometry	Page size and 1-inch margins
amsmath	align, \eqref, \dfrac, math environments
amssymb	Extra math symbols
graphicx	\includegraphics for the figures
booktabs	\toprule, \midrule, \bottomrule
listings	Code listing with syntax highlighting
xcolor	Colours for the code style
hyperref	Clickable cross-references and citations

Table 1: The eight packages used, top to bottom of the preamble.

The exact preamble lines:

```
\usepackage[a4paper, margin=1in]{geometry}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{graphicx}
\usepackage{booktabs}
\usepackage{listings}
\usepackage{xcolor}
\usepackage[hidelinks]{hyperref}
```

2 Document Structure

The title block is built from three commands in the preamble, then printed by one command in the body.

```
\title{A Concise Introduction to the Laplace Transform}
\author{Your Name \\\texttt{Your Student Id}}
\date{\today}
...
\begin{document}
\maketitle
\begin{abstract} ... \end{abstract}
```

2.1 `\title`, `\author`, `\date`

These three go in the *preamble* (before `\begin{document}`). They only *store* the information — nothing appears on the page until `\maketitle` is called inside the document. Leaving one out is fine: an omitted `\date` defaults to today, and an omitted `\author` just prints no author.

2.2 Line breaks and multiple authors

Inside `\author{}`, the double backslash `\\` forces a line break — that is how the student ID is pushed onto a second line beneath the name:

```
\author{Your Name \\ \texttt{Your Student Id}}
```

For *several* authors, separate them with `\and`, which spreads them across the line:

```
\author{Alice Rahman \\ \texttt{1905001}
\and
Bob Karim \\ \texttt{1905002}}
```

In the original article, the authors carried superscript numbers (Laplace^1) tied to affiliation lines below. That effect uses `\thanks` or manual $\text{\$}^{\text{\$1}}\text{\$}$ superscripts inside `\author{}`:

```
\author{Pierre-Simon Laplace\textsuperscript{\$1}\$ \\
\textsuperscript{\$1}\$Acad\'emie des Sciences, Paris}
```

2.3 `\maketitle` and abstract

`\maketitle` is what actually typesets the stored title, author, and date — it goes right after `\begin{document}`. The `abstract` environment (built into the `article` class) prints a centered “Abstract” heading followed by the indented summary paragraph; it is placed immediately after `\maketitle`.

2.4 A note on `acmart`

All of the above is standard `article`-class behaviour. In the `acmart` class (used for the CSE 200 paper assignment) the same commands exist but take extra structure — `\author` is paired with `\affiliation{}` and `\email{}` blocks, and the abstract must come *before* `\maketitle`, not after. So this title block would need rewriting to move to ACM format.

3 Sectioning

```
\section{Introduction}
\section{Definition}
\section{Key Properties}
```

Numbered automatically. A star (`\section*{}`) would remove the number.

4 Mathematics (`amsmath`)

4.1 Numbered single equation

```
\begin{equation}
F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt,
\label{eq:definition}
\end{equation}
```

Key pieces: `\mathcal{L}` for the script L; `\int_{~}^{~}` for the integral bounds; `\,` for a thin space; `\label{}` so it can be referenced.

→ *Rendered result:*

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t) e^{-st} dt$$

4.2 Aligned multi-line derivation

```
\begin{align}
RC\, s\, V_C(s) + V_C(s) &= \dfrac{V_{in}}{s}, \\
V_C(s) &= \dfrac{V_{in}}{s(RCs + 1)}.
\end{align}
```

The `&` sets the alignment point (before each `=`); `\\` ends each line. `align` numbers every line — use `align*` to suppress numbering.

→ *Rendered result:*

$$RC\, s\, V_C(s) + V_C(s) = \frac{V_{in}}{s},$$

$$V_C(s) = \frac{V_{in}}{s(RCs + 1)}.$$

4.3 Other math commands used

Command	Purpose
<code>\mathcal{L}</code>	Script letter for the transform
<code>\frac{~}{~}</code> / <code>\dfrac{~}{~}</code>	Fractions (dfrac = display size)
<code>\int_{~}^{~}</code>	Definite integral with bounds
<code>\infty</code>	Infinity symbol
<code>\tau</code> , <code>\omega</code> , <code>\sigma</code>	Greek letters
<code>\geq</code>	Greater-than-or-equal
<code>\left(\right)</code>	Auto-sized brackets
<code>\$V_{in}\$</code>	Subscript
<code>e^{-t/\tau}</code>	Superscript

Table 2: Inline and display math commands used in the solution.

=====

5 Code Listing (listings + xcolor)

Displaying source code uses two packages together: `listings` does the typesetting (the box, line numbers, keyword detection) and `xcolor` supplies the colours it refers to. The workflow is: (1) define colours, (2) bundle them into a named *style*, (3) apply that style when inserting code.

5.1 Step 1: Defining colours

```
\definecolor{codegreen}{rgb}{0.0,0.5,0.0}
\definecolor{codegray}{rgb}{0.5,0.5,0.5}
\definecolor{codepurple}{rgb}{0.58,0.0,0.82}
\definecolor{backcolour}{rgb}{0.96,0.96,0.96}
```

`\definecolor{name}{rgb}{r,g,b}` creates a named colour from red/green/blue values between 0 and 1. These names (`codegreen`, `backcolour`, ...) are then referred to in the style below. You could also use `{HTML}{FF5733}` for hex values.

5.2 Step 2: Defining the style

```
\lstdefinestyle{pystyle}{
  backgroundcolor=\color{backcolour},
  commentstyle=\color{codegreen},
  keywordstyle=\color{blue},
  stringstyle=\color{codepurple},
  basicstyle=\ttfamily\footnotesize,
  breaklines=true,
  numbers=left,
  numbersep=8pt,
  numberstyle=\tiny\color{codegray},
  tabsize=4,
  frame=single,
  language=Python
}
```

Each option, line by line:

Option	Effect
<code>backgroundcolor</code>	Fill colour behind the code
<code>commentstyle</code>	Colour of <code># comments</code> and docstrings
<code>keywordstyle</code>	Colour of language keywords (<code>def</code> , <code>for</code> , <code>if</code>)
<code>stringstyle</code>	Colour of "string literals"
<code>basicstyle</code>	Base font — monospace, small size
<code>breaklines</code>	Wrap long lines instead of overflowing
<code>numbers=left</code>	Put line numbers in the left margin
<code>numbersep</code>	Gap between numbers and code
<code>numberstyle</code>	Font/colour of the line numbers
<code>tabsize</code>	How many spaces a tab expands to
<code>frame=single</code>	Draw a box around the listing
<code>language</code>	Which language's keywords to highlight

Table 3: The `lstdefinestyle` options used in the solution.

`\lstdefinestyle{name}{...}` bundles all of this under one name (`pystyle`) so it can be reused without repeating the options.

5.3 Step 3a: Inserting an external file

```
\lstinputlisting[
  style=pystyle,
```

```
caption={Python routine for the RC step response.},
label={lst:rc}
]{rc_response.py}
```

`\lstinputlisting` reads the actual `.py` file from disk — so the code shown always matches the real source, and you never copy-paste it by hand. `caption` names the listing and `label` lets `\ref{lst:rc}` point at it. This is the form used in the solution.

→ *Rendered result:*

Listing 1: Python routine for the RC step response.

5.4 Step 3b: Typing code inline (alternative)

If you would rather write the code directly in the `.tex` file instead of loading an external file, use the `lstlisting` environment:

```
\begin{lstlisting}[style=pystyle, caption={...}]
def rc_response(R, C, Vin, times):
    return [...]
\end{lstlisting}
```

Everything between the tags is printed verbatim, so special characters like `_`, `&`, and `\` need no escaping inside it. Both forms accept the same options.

5.5 Why two files?

The listing package treats code as data. Whether it comes from an external `.py` (via `\lstinputlisting`) or is typed inline (via `lstlisting`), the `listings` package handles layout and line numbers, while `xcolor` provides every colour the style refers to. Remove `xcolor` and the `\color{...}` calls in the style break.

6 Cross-references and Citations (hyperref)

Command	Refers to
<code>\ref{fig:circuit}</code>	A figure number
<code>\ref{tab:pairs}</code>	A table number
<code>\ref{lst:rc}</code>	A listing number
<code>\eqref{eq:ode}</code>	An equation number, in parentheses
<code>\cite{ogata2010}</code>	A bibliography entry

Table 4: Reference commands. Each needs a matching `\label` (or `.bib` key for `\cite`).

`\eqref` is from `amsmath` and adds the parentheses automatically — (6) rather than 6. `hyperref` makes all of these clickable in the PDF.

7 Bibliography (BibTeX)

7.1 In the document

```
\bibliographystyle{plain}
\bibliography{laplace}
```

7.2 In laplace.bib

```
@book{kreyszig2011,  
  author = {Erwin Kreyszig},  
  title = {Advanced Engineering Mathematics},  
  edition = {10},  
  publisher = {John Wiley \& Sons},  
  year = {2011}  
}
```

`\cite{kreyszig2011}` then produces [1]. Compile order: `pdflatex` $\rightarrow$ `bibtex` $\rightarrow$ `pdflatex` $\rightarrow$ `pdflatex`.

8 Compile Order Reminder

1. `pdflatex main` — writes `.aux` with label and citation keys (references show as [?] and ??).
2. `bibtex main` — reads the `.aux`, builds the bibliography from `laplace.bib`.
3. `pdflatex main` — pulls the bibliography in.
4. `pdflatex main` — resolves all cross-reference numbers.

On Overleaf this all happens automatically on Recompile.

9 Title Block with Numbered Affiliations

The original article’s title carries superscript numbers on each author, tied to institution lines below (Laplace¹, Paris; Heaviside², London; ...). The cleanest way to get this is `\thanks`, which auto-numbers each note and prints the matching affiliation as a footnote.

```
\title{Laplace Analysis and Applications}  
\author{  
  Pierre-Simon Laplace\thanks{Acad\'emie des Sciences, Paris, France}  
\and  
  Oliver Heaviside\thanks{Royal Society, London, United Kingdom} \and  
  Gustav Robert Kirchhoff\thanks{University of Heidelberg, Germany}  
}  
\date{ }  
...  
\maketitle
```

How it works: `\thanks{...}` attaches a numbered footnote mark to whatever precedes it and prints its text at the bottom of the page — so each author automatically gets ¹, ², ³ with the institution beneath. `\and` separates authors on the same line. `\date{ }` (empty) removes the date.

Alternative — **affiliations stacked directly under the names** (matching the exact look of the article, where the three institutions sit as three centered lines rather than footnotes): use manual superscripts and `\\` inside `\author`:

```

\author{
  Pierre-Simon Laplace{1}$, Oliver Heaviside{2}$, and Gustav Robert
  Kirchhoff{3}$ \\\[1ex]
  {1}$Acad\'emie des Sciences, Paris, France \\\
  {2}$Royal Society, London, United Kingdom \\\
  {3}$University of Heidelberg, Heidelberg, Germany
}

```

Here 1 is a manual math-mode superscript, and `\\` breaks each affiliation onto its own centered line. This gives the three-line block exactly as it appears in the article, without footnotes.

→ *Rendered result:*

Laplace Analysis and Applications

Pierre-Simon Laplace¹, Oliver Heaviside², and Gustav Robert Kirchhoff³

¹Académie des Sciences, Paris, France

²Royal Society, London, United Kingdom

³University of Heidelberg, Heidelberg, Germany

10 Side-by-Side Text and Figure

On page 2 of the article, the derivation runs down the left while the circuit figure and the aligned right-hand side (Equation 6) sit to the right. Two `minipage` boxes placed next to each other produce this.

```

\noindent
\begin{minipage}[t]{0.52\textwidth}
  For a resistor  $R$  and capacitor  $C$ , ...
  \begin{equation}
    RC\frac{dv_C(t)}{dt} + v_C(t) = v_{in}(t),
  \end{equation}
  Assuming  $v_C(0) = 0$ , ...
  \begin{align*}
    & RC\left[sV_C(s) - v_C(0)\right] + V_C(s) \\\
    & \rightarrow RCsV_C(s) + V_C(s) \\\
    & \rightarrow V_C(s)(RCs + 1)
  \end{align*}
\end{minipage}
\hfill
\begin{minipage}[t]{0.42\textwidth}
  \centering
  \includegraphics[width=\textwidth]{rc_circuit.png}
  \captionof{figure}{A simple series RC circuit.}
  \begin{align}
    &= V_{in}(s), \text{ \notag } \\\
    &= V_{in}(s), \text{ \notag } \\\
    &= V_{in}(s).
  \end{align}
\end{minipage}

```

Key points:

- `\begin{minipage}[t]{0.52\textwidth}` makes a box 52% of the text width, top-aligned ([t]) so both columns start at the same height.
- `\hfill` between the two minipages pushes them to opposite margins.
- The two widths (0.52 and 0.42) sum to less than 1 to leave a gap; going over 1.0 makes the second box drop below the first.
- `\notag` on the first two `align` lines suppresses their numbers so only the final line is numbered (6), matching the article.
- `\captionof{figure}{...}` (from the `caption` package) is needed because a figure caption inside a `minipage` isn't in a `figure` float — add `\usepackage{caption}` for it.

→ *Rendered result:*

For a resistor R and capacitor C , Kirchhoff's voltage law gives

$$RC \frac{dv_C(t)}{dt} + v_C(t) = v_{in}(t),$$

Assuming $v_C(0) = 0$, taking the Laplace transform gives

$$\begin{aligned} RC [sV_C(s) - v_C(0)] + V_C(s) \\ \Rightarrow RCsV_C(s) + V_C(s) \\ \Rightarrow V_C(s)(RCs + 1) \end{aligned}$$

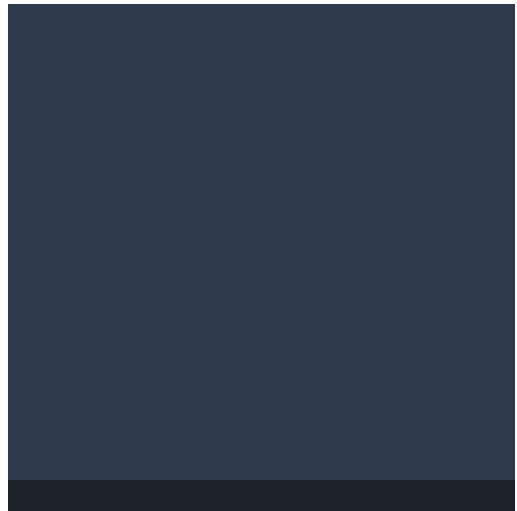


Figure: A simple series RC circuit.

$$\begin{aligned} &= V_{in}(s), \\ &= V_{in}(s), \\ &= V_{in}(s). \end{aligned}$$